Tuan Van Vo

Phone: +971504383972, Email: vovantuan1999a@gmail.com, Github: vovantuan1999a,
WebPage: https://vovantuan1999a.github.io/tuanvovan/

Education

Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI) Abu Dhabi, UAE
PhD in Computer Vision, supervised by Prof. Ian Reid *Sep. 2025 – Sep. 2029 (Expected)*

- MBZUAI is globally **ranked 10th** in AI, computational biology, computer vision, machine learning, natural language processing, and robotics according to CSRankings.

Vietnam National University Ho Chi Minh City University of Information Technology
Bachelor of Science in Computer Science *Aug. 2017 – June 2021*

- First Class with Honors **(Talented Class – Honors Program)**
- GPA: 8.32/10
- **Honors programs** offer special curricula, privileges, scholarships, and recognition for exceptional undergraduate students.

Publications and Under Submissions [Google Scholar]

Publications

[7] Duy Le Dinh Anh, Patrick Amadeus Irawan, **Tuan Van Vo**. Counting to Four is still a Chore for VLMs. In *CVPR 2026 Workshops – The 5th Workshop on Multimodal Foundation Models*, 2026. Paper

[6] Tudor Jianu, Baoru Huang, **Tuan Van Vo**, ..., Olatunji Omisore, Pierre Berthet-Rayne, Sebastiano Fichera, Anh Nguyen. Autonomous Catheterization with Open-source Simulator and Expert Trajectory. In *Handbook of Robotic and Image-Guided Surgery*, 2026. Paper Code

[5] Sy-Tuyen Ho*, **Tuan Van Vo***, Somayeh Ebrahimkhani*, Ngai-Man Cheung. Vision Transformer Neural Architecture Search for Out-of-Distribution Generalization: Benchmark and Insights. In *Conference on Neural Information Processing Systems (NeurIPS)*, 2024. Paper Code. ***First authors.**

[4] **Tuan Van Vo**, Minh Nhat Vu, Baoru Huang, Toan Nguyen, Ngan Le, Thieu Vo, Anh Nguyen. Open-Vocabulary Affordance Detection using Knowledge Distillation and Text-Point Correlation. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2024. **Oral Presentation.** Paper Code Video

[3] Toan Nguyen, Minh Nhat Vu, Baoru Huang, **Tuan Van Vo**, Thuy Tuong Vy Truong, Ngan Le, Thieu Vo, Hoai Bac Le, Anh Nguyen. Language-Conditioned Affordance-Pose Detection in 3D Point Clouds. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2024. **Oral Presentation.** Paper Code Video

[2] **Tuan Van Vo**, Minh Nhat Vu, Baoru Huang, An Vuong, Ngan Le, Thieu Vo, Anh Nguyen. Language-driven Grasp Detection with Mask-guided Attention. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024. **Oral Presentation.** Paper Code

[1] Nam Phuong Nguyen*, **Tuan Van Vo***, Soan T. M. Duong, Chanh D. Tr. Nguyen, Trung Bui, Steven Q. H. Truong. Dual Consistency Assisted Multi-Confident Learning for Hepatic Vessel Segmentation using Noisy Labels. In *British Machine Vision Conference (BMVC)*, 2022. Paper Code. ***First authors.**

Preprints

[10] An Dinh Vuong*, **Tuan Van Vo***, Abdullah Sohail, Haoran Ding, Liang Ma, Xiaodan Liang, Anqing Duan, Ivan Laptev, Ian Reid. World2Act: Latent Action Post-Training via Skill-Compositional World Models. *Under review*, 2026. arXiv Project Page. ***First authors.**

[9] **Tuan Van Vo**, Quang Tan Nguyen, Khang Nguyen, Tran Xuan Nhat, Duy Minh Ho Nguyen, An Thai Le, Vien Anh Ngo, Minh Nhat Vu. ReFineVLA: Multimodal Reasoning-Aware Generalist Robotic Policies via Teacher-Guided Fine-Tuning. *Under review*, 2025. arXiv

[8] Baoru Huang*, **Tuan Van Vo***, Chayun Kongtongvattana, Anh Nguyen. CathAction: A Benchmark for Endovascular Intervention Understanding. *Under review*, 2024. arXiv. ***First authors.**

Academic Services

Reviewer Services - Journal and Conference Reviewer

- Reviewer for TMM-24, ICRA-25, IROS-25, CoRL-25, CoRL-26

Work Experience

Research Lab Experience

Singapore University of Technology and Design, Singapore Sept. 2023 – Mar. 2025

Research Assistant

- Research focus: Zero-Shot Neural Architecture Search
- Vision Transformer Neural Architecture Search for Out-of-Distribution Generalization, **resulting in a publication at NeurIPS 2024**

FPT Software AI Center July. 2022 – June. 2024

AI Research Residency

- Research focus: Multimodal Robot Learning
- **Resulted in two publications at ICRA 2024 and one at IROS 2024**

MMLab - UIT Oct. 2019 – May 2021

Research Assistant, *University of Information Technology (UIT) - VNUHCM*

- Research Thesis: Violence Detection in Video Surveillance Using a Deep Multiple Instance Learning Ranking Approach (PDF)

Industry Experience

VinRobotics, Vingroup Corporation April. 2025 – Aug. 2025

Principal Research Engineer (Founding Team Member) *Vietnam*

- Research focus: Foundation Models for Vision-Language-Action (VLA) in robotics.
- Lead scientist developing large-scale multimodal VLA models that integrate vision, natural language, and action planning for next-generation robotics at VinRobotics.

MoMo (M-Service JSC) June. 2022 – August. 2023

Senior Computer Vision Research Engineer

- Researched and developed AI solutions, specifically designing computer vision models for the eKYC system of the MoMo app
- Designed and implemented Face Anti-Spoofing, Object Detection, OCR, Deep-Fake Detection, Fraud Detection, and Document Understanding models
- Achieved a 99% accuracy rate with low latency, serving nearly 40 million users at about 45 RPS
- **These AI eKYC systems have gained trust and are utilized by the Apple Online Store in Vietnam**

VinBrain (Acquired by NVIDIA) May. 2021 – May. 2022

Applied Scientist

- Conducted research and development of AI for medical segmentation and diagnostic tasks, as well as AI for Smart City and Smart Home solutions
- Designed models for 3D MRI CT-Rectal Tumor Segmentation, Liver Vessel Segmentation, Pet Detection, and Coverface Detection on surveillance cameras
- Successfully delivered all AI models to products to assist doctors. In the surveillance camera project, the AI models were also successfully delivered and deployed at VinHome residential areas for camera-AI based surveillance
- Performed in-depth research and targeted academic papers for prestigious AI conferences, **resulting in a publication at BMVC 2022**

FPT Software May. 2020 – May. 2021

AI Engineer - Technical Lead of Healthcare Projects

- Designed and implemented cutting-edge AI solutions for healthcare in the Japanese market, addressing customer requirements related to model design, logic, evaluation, and reporting
- Designed models for semantic segmentation of dental X-rays and classification of 72 implant types
- Successfully delivered all AI models to the customer

Scholarship and Activities

Full Scholarship and Stipend for PhD Study at Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI)
Rosen Scholarship 2020 & 2021
UIT-Study Scholarships 2018 & 2019
Participated in the ICPC 2019 Vietnam National Programming Contest
Attended and presented two oral papers at ICRA 2024, Yokohama, Japan

Skills

Programming Languages: Python, C/C++, SQL
AI Frameworks and Model Training: PyTorch, TensorFlow, Keras, OpenCV, CUDA basics; experienced in large-scale distributed training of VLA, VLM, LLM, MLLM, Video Generation, and Video World models
Robotics and Deployment: ROS, robotic system deployment, teleoperation-based data collection
Tools and Platforms: Linux, Docker, Google Cloud, Jupyter Notebook/Lab, Git, BigQuery, Oracle
Languages: English, Vietnamese (native)
Typography: LaTeX

References

- **Prof. Ian Reid**
  Department Chair and Professor of Computer Vision at MBZUAI
  Acting Director of Research Administration at MBZUAI
  Email Ian.Reid@mbzuai.ac.ae
  Google Scholar https://scholar.google.com/citations?user=ATkNLcQAAAAJ&hl=en
  Homepage https://mbzuai.ac.ae/study/faculty/ian-reid/

- **Prof. Anh Nguyen**
  Associate Professor at the University of Liverpool
  Director of the Smart Robotic Lab, Department of Computer Science
  Email anh.nguyen@liverpool.ac.uk
  Google Scholar https://scholar.google.com/citations?user=gEbaF0sAAAAJ&hl=en
  Homepage https://www.csc.liv.ac.uk/~anguyen/

- **Prof. Ngai-Man (Man) Cheung**
  Associate Professor at the Singapore University of Technology and Design (SUTD)
  Associate Pillar Head (Education)
  Email ngaiman_cheung@sutd.edu.sg
  Google Scholar https://scholar.google.com/citations?user=vJQArFkAAAAJ&hl=en
  Homepage https://sites.google.com/site/mancheung0407/

- **Prof. Thieu N. Vo**
  Assistant Professor at the University of Bath
  Department of Computer Science.
  Email ntv22@bath.ac.uk
  Google Scholar https://scholar.google.com/citations?user=CM2qJSoAAAAJ&hl=en

- **Prof. Baoru Huang**
  Assistant Professor at the University of Liverpool
  Department of Computer Science
  Email bh1518@ic.ac.uk
  Google Scholar https://scholar.google.com/citations?user=unbPvWAAAAAJ&hl=zh-EN
  Homepage https://baoru.netlify.app/